KisaanMadad

Project Proposal

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Abstract

Small-scale farmers in Pakistan struggle with limited access to market intelligence and scientific farming practices, resulting in suboptimal decisions and reduced profitability. KisaanMadad addresses these challenges through an integrated mobile decision support system combining machine learning and agricultural science.

The system provides probabilistic price forecasting using historical mandi data, enabling informed planting decisions. A multi-criteria crop selection module recommends optimal crops based on weather patterns, market proximity, and planting schedules. The irrigation component implements FAO-56 methodology to calculate crop-specific water requirements, considering weather forecasts, soil moisture, growth stages, and water costs. Financial planning tools include loan calculators, revenue projections, and expense tracking.

Developed using Pakistani agricultural datasets, KisaanMadad bridges the information gap for resource-constrained farmers, offering end-to-end support from pre-planting to post-harvest financial analysis.

1. Introduction

Agriculture remains the backbone of Pakistan's economy, contributing 23.54% to the national GDP and employing 37.4% of the labor force [1]. Despite this significant role, small and medium-scale farmers—who now constitute 97% of the farming community yet typically own less than 12.5 acres each [2]—face persistent challenges that limit their productivity and profitability. Research indicates a yield gap of 40-60% for major crops like wheat and sugarcane when comparing average farm yields to the potential demonstrated in progressive farming and research trials [3].

The challenges are multifaceted and severe. Price volatility is intensified by macroeconomic instability and strong spillover effects between major commodities like wheat and rice [4], yet farmers lack access to reliable forecasting tools, forcing them to make planting decisions blindly. Water scarcity presents another critical constraint: Pakistan is among the world's most water-stressed countries, with per capita water availability declining from 5,260 cubic meters in 1951 to just 908 cubic meters in 2021—well below the international scarcity threshold of 1,000 cubic meters [5]. Despite this crisis, irrigation efficiency remains below 40% [6], with farmers over-irrigating some crops while under-irrigating others due to lack of scientific guidance.

KisaanMadad addresses these interconnected challenges through an integrated mobile decision support system combining machine learning and agricultural science. The system provides four core capabilities: (1) probabilistic price forecasting using historical mandi data to help farmers navigate market volatility, (2) multi-criteria crop recommendations based on weather patterns, market proximity, and planting schedules, (3) FAO-56 compliant irrigation scheduling that calculates crop-specific water requirements to optimize scarce water resources, and (4) financial planning tools including loan calculators, revenue projections, and expense tracking to improve access to credit and financial management.

By consolidating these capabilities into a platform designed for the Pakistani agricultural context, KisaanMadad aims to bridge critical information gaps, improve resource efficiency, and enhance farmer welfare in a sector vital to national food security and economic stability.

2. Goals and Objectives

2.1 Primary Goal

To develop an intelligent decision support system that empowers Pakistani farmers with data-driven insights for improved agricultural decision-making, resource optimization, and financial planning.

2.2 Specific Objectives

Our project's objectives include:

- **Researching existing agricultural decision support systems** and identifying their limitations in the Pakistani context, particularly regarding data accessibility and integration for small-scale farmers.
- **Collecting and preprocessing agricultural datasets** from Pakistani sources including mandi prices, weather data, crop coefficients, and soil properties.
- **Developing a probabilistic price forecasting model** that analyzes historical market data to provide farmers with risk-aware market intelligence for crop selection and timing decisions.
- **Implementing a multi-criteria crop recommendation system** that evaluates weather, market proximity, water availability, and planting schedules to suggest optimal crops.
- **Designing an irrigation scheduling module** based on FAO-56 methodology that calculates crop-specific water requirements using weather forecasts, soil moisture simulation, and growth stage data.
- **Creating financial planning tools** including loan calculators, revenue projections, and expense trackers for comprehensive financial management.
- **Identifying optimal machine learning models** that balance accuracy and computational efficiency for mobile deployment.
- **Validating system recommendations** through backtesting on historical data and evaluation against actual farming outcomes.
- **Deploying an intuitive mobile application** with Urdu language support accessible to farmers with varying digital literacy levels.

3. Scope of the Project

This project develops an agricultural decision support system for small and medium-scale farmers in Punjab, Pakistan, focusing on major crops including wheat, cotton, and rice. The system addresses four key areas: price forecasting, crop selection, irrigation scheduling, and financial planning.

Historical mandi price data, weather information, and agricultural datasets will be collected from publicly available Pakistani sources. Machine learning models will provide probabilistic price forecasts and multi-criteria crop recommendations. The irrigation module will implement FAO-56 methodology to calculate crop-specific water requirements based on weather forecasts and growth stages.

The system will be deployed as a mobile application with Urdu language support and an intuitive interface accessible to farmers with varying digital literacy levels. Validation will be conducted through backtesting on historical agricultural data from Punjab province.

4. Initial Study and Work Done so Far

Agricultural decision support systems have evolved significantly, yet their adoption in developing countries remains limited. Research indicates that while precision agriculture tools exist in developed markets, they rely on expensive sensor infrastructure and assume high-speed internet connectivity—constraints that make them inaccessible to smallholder farmers in Pakistan [7]. Studies examining agricultural price forecasting highlights distinct trade-offs: while Prophet offers ease of tuning and ARIMA captures linear trends effectively, hybrid Deep Learning models (CNN-LSTM) deliver the highest predictive fidelity. By benchmarking against these standard models, advanced systems can leverage specific architectures to minimize error rates, which have been shown to improve by up to 23% over traditional statistical methods [8]. However, most research focuses on global commodity exchanges rather than local wholesale markets (*mandis*) relevant to Pakistani farmers.

The FAO-56 Penman-Monteith methodology remains the international gold standard for irrigation scheduling, having been validated globally to consistently outperform other estimation methods in diverse climatic conditions [9]. Mobile applications implementing simplified ET_0 calculations, such as the Hargreaves equation, have shown promise for regions with limited weather data availability [10]. Despite these advances, existing Pakistani agriculture apps remain fragmented by offering either market prices or weather information, but rarely integrating multiple decision support features within a unified platform [7].

Our preliminary work has focused on establishing a robust data foundation and researching the critical pitfalls within Pakistan's agricultural ecosystem. We have identified and evaluated the primary datasets necessary for the KisaanMadad decision support algorithms, including historical daily price data (2018–2023) for major commodities like wheat, cotton, and rice from the Punjab Agriculture Marketing Regulatory Authority (PAMRA) and Agriculture Marketing Information Service (AMIS). Additionally, we validated the NASA POWER API for agro-climatology inputs—such as solar radiation, precipitation, and temperature—and integrated FAO-56 standards for crop coefficients (K_c) to ensure scientifically grounded irrigation planning. Simultaneously, our research into local farming challenges highlighted severe issues such as water mismanagement, where canal losses reach nearly 38% and per capita availability has dropped below 1,000 m^3, as well as market asymmetry that forces smallholder farmers to sell to middlemen (*arthies*) at undervalue rates. This groundwork confirms the feasibility of a data-driven approach and frames the specific problems KisaanMadad is designed to solve.

Critically, our review revealed that financial planning tools are notably absent from existing solutions despite being essential for farmer decision-making. The integration of loan calculators, revenue projections, and expense tracking alongside agronomic recommendations represents a novel contribution that addresses the holistic needs of resource-constrained farmers.

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